

Principles:

The American Chemical Society (ACS) recognizes the importance of artificial intelligence as an emerging technology that has significant potential to create change. ACS is excited for the potential AI provides as a tool for innovation, research, and other improvements to society. AI has been used extensively and increasingly in the chemical sciences in applications such as predicting the folding of proteins, developing various drug discovery systems, and predicting chemical properties. ACS also recognizes the critical importance that AI systems be developed and deployed safely and responsibly to avoid causing harm to human life, health, or the environment, and to ensure respect for intellectual property rights. Regulations on AI will affect the chemical enterprise.

ACS believes the following principles are key when developing, using, deploying, and regulating artificial intelligence:

- Privacy and Security
 - Individual's data is handled responsibly, including the ability to opt-out or opt-in to data use in AI, depending on the use-case.
 - Sensitive information is handled responsibly and securely during training and use of AI.
 - Established rights should be protected by shielding individuals against discrimination, fraud, and privacy invasions that emerge from AI use.
- Transparency and Disclosure
 - AI systems should be trained on content and data which is accessed lawfully and, where the intellectual property of others is used, prior authorization has been obtained.
 - The content and sources used to train an AI system should be publicly and clearly identified.
 - Users of AI based tools are made aware of the presence and use of AI in those interactions.
 - Readers and users of AI-generated materials are made aware of how AI was used in the material's creation.
 - AI actors should be held accountable for the proper functioning of AI systems, the model's generated output, and for respecting intellectual property rights and the privacy and dignity of all individuals involved.
- Education and Workforce Development
 - AI literacy should be promoted to equip individuals to engage productively and responsibly with AI technologies both personally and professionally.
 - AI systems should be used to help all students achieve their educational goals while still fostering learning skills, particularly considering the ongoing digital divide, and educators should have workforce support and retraining to effectively use AI tools.

- AI systems used in education must ensure privacy, data security, student safety, and child and youth protection. Data collection and monitoring, data transfer and ownership policies should be specific and transparent.
- Decision-making functions supported by AI must allow for human intervention and ultimately rely on human approval processes. AI systems should serve in supportive roles without replacing the responsibilities of students, teachers, hiring professionals or administrators. Mechanisms should be in place to override, repair, or decommission AI systems at risk of causing undue harm or exhibiting undesired behavior, and appeals processes should be established.
- Social Impacts
 - AI use and deployment should not operate to reinforce inequality.
 - AI systems should promote fairness and seek to avoid causing disproportionate harm to any person or group of people.
 - Developers and regulators should work to intentionally engage multiple stakeholders when developing and regulating AI.
 - AI technologies should be designed and deployed in a way which is not harmful to the environment or to human life, health or dignity.
- Evaluation
 - Guidance for all aspects of AI should be reviewed and updated regularly to ensure compliance and relevance with rapidly changing laws, regulations, and technology.
 - As AI infrastructure increases energy usage, grid resiliency and flexibility, use of a variety of energy sources, efficiency and sustainability of the energy used, as well as security must be considered. The prioritization of long-term sustainability and resiliency will help create an AI infrastructure that allows the US to remain a leader in technology and innovation.
- Innovation
 - AI technology is expensive, and dependent on large data sets and computing power, giving an advantage to large companies. Infrastructure required for the effective development of new AI technologies should be accessible to universities and companies of all sizes.
 - As well as computing power and technology, non-proprietary content and data required to develop novel models and algorithms should be widely available and of high quality. Wherever possible, the government should support efforts to develop databases of these data that are accessible to the public.

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